Appendix P
THE EXCLUSION PRINCIPLE IN *LS* COUPLING

If an atom contains two or more electrons that have common values of the quantum numbers n and l, because they are in the same subshell, the exclusion principle imposes restrictions on the possible values of the remaining quantum numbers. In the Hartree approximation, these are the m_l and m_s quantum numbers of each electron. In this case the exclusion principle says simply that no two electrons can have the same set of all four quantum numbers. In LS coupling, the quantum numbers that are used, in addition to n and l for each electron, are l', s', j', m_j'. These quantum numbers specify the way the electrons interact in LS coupling. The restrictions imposed by the exclusion principle on the possible values of these quantum numbers are more complicated, but they can be determined as follows.

Working first in the Hartree approximation, the possible values of m_l and m_s are used to determine the possible values of the quantum numbers m_l', m_s', m_j'. From these the possible values of l', s', j', m_j' are then determined. Although in LS coupling the z components of $\mathbf{L}'$ and $\mathbf{S}'$, which are specified by m_l' and m_s', are changed by the residual Coulomb and spin-orbit interactions, L', S', J', J_z' are not changed. Therefore, the restrictions that are found in the Hartree approximation concerning the associated quantum numbers, l', s', j', m_j', also apply in LS coupling.

As an example, we determine the LS coupling quantum numbers which satisfy the exclusion principle for two electrons in the $2p$ subshell. Referring to Table P-1, we first list all the possible sets of values of m_l and m_s for the two electrons, which satisfy the exclusion principle. There are 15 different sets of m_l and m_s for the two electrons which satisfy the exclusion principle, and a number of others, such as $m_{l_1} = +1$, $m_{s_1} = +1/2$, $m_{l_2} = +1$, $m_{s_2} = +1/2$,

Table P-1 Possible Quantum Numbers for an np^2 Configuration

Entry	m_{l_1}	m_{s_1}	m_{l_2}	m_{s_2}	m_l'	m_s'	m_j'
1	+1	+1/2	+1	−1/2	+2	0	+2
2	+1	+1/2	0	+1/2	+1	+1	+2
3	+1	+1/2	0	−1/2	+1	0	+1
4	+1	+1/2	−1	+1/2	0	+1	+1
5	+1	+1/2	−1	−1/2	0	0	0
6	+1	−1/2	0	−1/2	+1	−1	0
7	+1	−1/2	−1	+1/2	0	0	0
8	+1	−1/2	−1	−1/2	0	−1	−1
9	0	+1/2	+1	−1/2	+1	0	+1
10	0	+1/2	0	−1/2	0	0	0
11	0	+1/2	−1	+1/2	−1	+1	0
12	0	+1/2	−1	−1/2	−1	0	−1
13	−1	+1/2	0	−1/2	−1	0	−1
14	−1	+1/2	−1	−1/2	−2	0	−2
15	−1	−1/2	0	−1/2	−1	−1	−2

Table P-2 Possible Quantum Numbers for an np^6 Configuration

Entry	m_{l_1}	m_{s_1}	m_{l_2}	m_{s_2}	m_{l_3}	m_{s_3}	m_{l_4}	m_{s_4}	m_{l_5}	m_{s_5}	m_{l_6}	m_{s_6}	m_l'	m_s'	m_j'
1	$+1$	$+1/2$	$+1$	$-1/2$	0	$+1/2$	0	$-1/2$	-1	$+1/2$	-1	$-1/2$	0	0	0

which are ruled out because they violate it. For each set the corresponding values of the quantum numbers m_l', m_s', m_j' are evaluated from the relations $m_l' = m_{l_1} + m_{l_2}$, $m_s' = m_{s_1} + m_{s_2}$, $m_j' = m_l' + m_s'$, which represent z components of the angular momentum addition equations, (10-6), (10-8), and (10-10).

The problem now is to identify the allowed quantum states, specified in Table P-1 in terms of m_l', m_s', m_j', with the specification of these states in terms of l', s', j'. We begin by using (10-14), which represent other requirements of angular momentum conservation. Setting $l_1 = l_2 = 1$, we find that the possible combinations of l', s', j', expressed in spectroscopic notation, are as follows: ${}^1S_0$, ${}^1P_1$, ${}^1D_2$, ${}^3S_1$, ${}^3P_0$, ${}^3P_1$, ${}^3P_2$, ${}^3D_1$, ${}^3D_2$, ${}^3D_3$. The ${}^3D_3$ states are immediately ruled out because for these states there would be m_j' values of $+3$ and -3, but we see that there are none listed in Table P-1. Since there are no ${}^3D_3$ states, there can be no ${}^3D_2$ or ${}^3D_1$ states; all these states correspond to $\mathbf{S}'$ and $\mathbf{L}'$ vectors of the same magnitude in the same multiplet and they stand or fall together. Now, entry number 1 in the table says there must be states with $s' \geq 0$ and $l' \geq 2$, since $m_s' = -s', \ldots, s'$ and $m_l' = -l', \ldots, l'$. These requirements can be satisfied only by the states ${}^1D_2$. There are five such states corresponding to the five values $m_j' = -2, -1, 0, 1, 2$. Entry number 2 says that there must be states with $s' \geq 1$ and $l' \geq 1$. This requires the presence of the states ${}^3P_0$, ${}^3P_1$, ${}^3P_2$. For ${}^3P_0$ there is one state corresponding to $m_j' = 0$. For ${}^3P_1$ there are three states corresponding to $m_j' = -1, 0, 1$. For ${}^3P_2$ there are five corresponding to $m_j' = -2, -1, 0, 1, 2$. The number of states we have identified so far is $5 + 1 + 3 + 5 = 14$. Only a single state is left, and this must be a state with $m_j' = 0$ because all the other m_j' values of the table have been used. It is clear then that this must be the single quantum state ${}^1S_0$.

We have found that in the Hartree approximation the only possible quantum states for two electrons with the configuration $2p^2$ are those associated with the symbols ${}^1S_0$, ${}^1D_2$, ${}^3P_{0,1,2}$. This is equally true for an np^2 configuration with any n. Since these restrictions are expressed in terms of the quantum numbers l', s', j', they are also valid in *LS* coupling. Note that these results agree with the states that are observed to be present in the ^{6}C energy-level diagram of Figure 10-8.

As a second example, consider six electrons in the same p subshell, that is, consider the configuration np^6, with any n. Table P-2 lists the allowed quantum states for this case, in analogy to the listing for the np^2 configuration, but in the present case the table has only one entry. The entry is obviously the single state ${}^1S_0$. Of course, six electrons represents the maximum number that can occupy a p subshell. Thus we conclude that when this subshell is filled, its total spin angular momentum, total orbital angular momentum, and total angular momentum, are all zero. Furthermore, it is apparent that the same conclusion will be obtained for any completely filled subshell. The conclusion is confirmed by the analysis of the optical spectra of noble gas atoms. Also, if a completely filled subshell has no net spin or orbital angular momentum, there can be no net magnetic dipole moment. This is confirmed by Stern-Gerlach experiments on noble gas atoms.

Table P-3 lists the quantum states allowed by the exclusion principle for some configurations containing several electrons in the same subshell. Each symbol gives the l' and s' values of an allowed multiplet. The possible values of j' and m_j' for the states of that multiplet can be determined in terms of l' and s' from (10-13) and (10-14). Entries are given for configurations ranging from no electrons in the subshell up to the maximum number of electrons consistent with the exclusion principle. For no electrons, $l' = s' = j' = 0$, which is described by the symbol ${}^1S_0$. For one electron in any subshell, $s' = 1/2$, and the allowed states are necessarily ${}^2S_{1/2}$, or ${}^2P_{1/2,3/2}$, etc. The allowed states for other configurations are determined by the calculations in the examples above, or by similar calculations. The allowed states can also be obtained from more elegant calculations based on the mathematical theory of groups.

It is particularly interesting to note the symmetries in Table P-3 about the half-filled subshell configurations. The number of states is greatest for this configuration, and the states for a configuration in which a subshell is filled except for a certain number of electrons are

Table P-3 Possible Quantum Numbers for Configurations Containing Several Electrons in the Same Subshell

ns^0	^1S					
ns^1		^2S				
ns^2	^1S					
np^0	^1S					
np^1		^2P				
np^2	^1S, ^1D		^3P			
np^3		^2P, ^2D		^4S		
np^4	^1S, ^1D		^3P			
np^5		^2P				
np^6	^1S					
nd^0	^1S					
nd^1		^2D				
nd^2	^1S, ^1D, ^1G		^3P, ^3F			
nd^3		^2D, ^2P, ^2D, ^2F, ^2G, ^2H		^4P, ^4F		
nd^4	^1S, ^1D, ^1G, ^1S, ^1D, ^1G, ^1F, ^1I		^3P, ^3F, ^3P, ^3D, ^3F, ^3G, ^3H		^5D	
nd^5		^2D, ^2P, ^2D, ^2F, ^2G, ^2H, ^2S, ^2D, ^2F, ^2G, ^2I		^4P, ^4F, ^4D, ^4G		^6S
nd^6	^1S, ^1D, ^1G, ^1S, ^1D, ^1G, ^1F, ^1I		^3P, ^3F, ^3P, ^3D, ^3F, ^3G, ^3H		^5D	
nd^7		^2D, ^2P, ^2D, ^2F, ^2G, ^2H		^4P, ^4F		
nd^8	^1S, ^1D, ^1G		^3P, ^3F			
nd^9		^2D				
nd^{10}	^1S					

exactly the same as the states for the configuration in which there are just that number of electrons in the subshell. This result can also be expressed by saying that the allowed states for electrons are the same as the allowed states for holes—a fact that has important consequences in solid state and nuclear physics, as well as atomic physics. The symmetries are a striking demonstration of the effect of the exclusion principle because, if it were not for this principle, the number of states would increase monotonically as the number of electrons in the subshell increased.